

Jan Janssen

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PROFESSIONAL APPOINTMENTS

- 2023 – present **Group leader for Materials Informatics** - Computational Materials Design
Research topic: foundation models for sustainable materials
Max Planck Institute for Sustainable Materials, Düsseldorf, Germany
- 2024 **Invited Fellow** - Institute for Mathematical and Statistical Innovation (IMSI)
Research topic: data-driven materials informatics
University of Chicago, Chicago, IL, USA
- 2022 – 2023 **Director's Postdoctoral Fellowship** - Theoretical Division (T-1)
Research topic: long-timescale simulations of materials with *ab initio* accuracy
Los Alamos National Laboratory, Los Alamos, NM, USA
- 2023 **Invited Postdoctoral Fellow** - Institute for Pure and Applied Mathematics (IPAM)
Research topic: new mathematics for the exascale: applications to materials science
University of California, Los Angeles, CA, USA
- 2021 – 2022 **Postdoctoral Research Associate** - Theoretical Division (T-1)
Research topic: exascale atomistic capability for accuracy, length, and time
Los Alamos National Laboratory, Los Alamos, NM, USA
- 2017 **Invited Fellow** - Institute for Pure and Applied Mathematics (IPAM)
Research topic: complex high-dimensional energy landscapes
University of California, Los Angeles, CA, USA
- 2015 – 2021 **PhD Candidate** - Computational Materials Design
Research topic: computational phase studies and *ab initio* thermodynamics
Max-Planck-Institut für Eisenforschung, Düsseldorf, Germany

EDUCATION

- 2015 – 2021 **PhD in Theoretical Physics** - Paderborn University, Germany
- Thesis: pyiron - an integrated development environment for *ab initio* thermodynamics
 - Advisor: Prof. Jörg Neugebauer
 - Grade: summa cum laude
- 2009 – 2015 **Advanced Degree in Physics** - Technical University of Kaiserslautern, Germany
- Thesis: carbon in α -iron-grainboundaries - an atomistic study of elastic properties
 - Advisor: Prof. Herbert Urbassek
 - Grade: thesis 1.0 (excellent) - total 1.6 (good)

FUNDED PROPOSALS

- 2024 **Principal Investigator** as part of the International Max Planck Research School for Sustainable Metallurgy - from Fundamentals to Engineering Materials
- Funding: EUR 150k
- Co-Principal Investigator** as part of the collaborative research center (CRC) 1394 “Structural and chemical atomic complexity - from defect phase diagrams to material properties“
- Funding: EUR 160k
- 2023 **Principal Investigator** of “workflows for machine learned interatomic potentials“ as part of an internal call for funding for Postdoctoral Fellows at Los Alamos National Laboratory
- Computing Hardware Funding: USD 13k
- Co-Principal Investigator** of “uncertainty propagation for multi-fidelity machine learned interatomic potentials“ as part of the summer school for applied machine learning at Los Alamos National Laboratory
- Funding: USD 25k
- 2022 **Co-Principal Investigator** of “development and deployment of a fully autonomous *in silico* processing and materials discovery platform“ in collaboration with the department of Mechanical Engineering at Texas A&M University
- Funding: USD 176k
- Co-Principal Investigator** of “transferability of interatomic machine learning potentials“ as part of the summer school for applied machine learning at Los Alamos National Laboratory
- Funding: USD 25k
- Principal Investigator** of “helping users build workflows with ontological typing“ project funded by a NumFOCUS small development grant to hire an independent contractor
- Funding: USD 6k

AWARDS & HONORS

- 2024 Winner of the “2nd Annual Large Language Model Hackathon for Applications in Materials Science and Chemistry“ coupling a large language model to atomistic simulation
- Invited fellow at the Institute for Statistical and Mathematical Innovation (IMSI)
- 2023 Invited postdoctoral fellow at the Institute for Pure and Applied Mathematics (IPAM)
- 2022 Los Alamos National Laboratory (LANL) director’s postdoctoral fellowship
- 2021 PhD with highest distinction - summa cum laude
- 2019 Runner-up for the Heinz Billing Award - a national biennial award for outstanding contributions to computational science by younger scientist without tenure
- 2017 Invited fellow at the Institute for Pure and Applied Mathematics (IPAM)
- 2015 Scholarship of the Technical University of Kaiserslautern

PUBLICATIONS

Google Scholar, 04/26: Citations: 501

- 2026 S. Brinckmann, S. Menon, R. Röske, J. Steinhülb, M. Selzer, **J. Janssen**, T. Hickel, R. Schwaiger and J. Neugebauer. A Lightweight Procedural Layer for Hybrid Experimental-Computational Workflows in Materials Science. *Adv. Eng. Mater.* doi:[10.1002/adem.202503176](https://doi.org/10.1002/adem.202503176). Citations: 0
- Sk Md A.A. Alvi, B. Vela, V. Attari, **J. Janssen**, D. Perez, D. Allaire, and R. Arróyave. Deep Gaussian process-based cost-aware batch Bayesian optimization for complex materials design campaigns. *npj Computational Materials*. doi:[10.1038/s41524-026-01981-7](https://doi.org/10.1038/s41524-026-01981-7). Citations: 2
- M. Stricker, L. Banko, N. Sarazin, N. Siemer, **J. Janssen**, L. Zhang, J. Neugebauer and A. Ludwig. Computationally accelerated experimental materials characterization - drawing inspiration from high-throughput simulation workflows. *npj Computational Materials*. doi:[10.1038/s41524-025-01919-5](https://doi.org/10.1038/s41524-025-01919-5). Citations: 6
- I. Baghishov, **J. Janssen**, G. Henkelman and D. Perez. Application-specific Machine-Learned Interatomic Potentials: Exploring the Trade-off Between DFT Convergence, MLIP Expressivity, and Computational Cost. *Digital Discovery*. doi:[10.1039/D5DD000294J](https://doi.org/10.1039/D5DD000294J). Citations: 1
- Code:  [IlgarBaghishov/DFTprecisionMLIP](https://github.com/IlgarBaghishov/DFTprecisionMLIP)
- 2025 Sk Md A.A. Alvi, M. Mulukutla, N. Flores, D. Khatamsaz, **J. Janssen**, D. Perez, D. Allaire, V. Attari and R. Arróyave. Accurate and uncertainty-aware multi-task prediction of HEA properties using prior-guided deep Gaussian processes. *npj Computational Materials*. doi:[10.1038/s41524-025-01811-2](https://doi.org/10.1038/s41524-025-01811-2). Citations: 2
- Code:  [sheikhahnaf/DGP_with_EncoderDecoderPrior](https://github.com/sheikhahnaf/DGP_with_EncoderDecoderPrior)
- J. Janssen**, J. George, J. Geiger, M. Berx, X. Wang, C. Ertural, J. Schaarschmidt, A.M. Ganose, G. Pizzi, T. Hickel and J. Neugebauer. A Python workflow definition for computational materials design. *Digital Discovery*. doi:[10.1039/D5DD000231A](https://doi.org/10.1039/D5DD000231A). Citations: 5
- Code:  [pythonworkflow/python-workflow-definition](https://github.com/pythonworkflow/python-workflow-definition)
- Y. Zimmermann, A. Bazgir, A. Al-Feghali, M. Ansari, L.C. Brinson, Y. Chiang, D. Circi, M.-H. Chiu, N. Daelman, M.L. Evans, A.S. Gangan, J. George, H. Harb, G. Khalighinejad, S.T. Khan, S. Klawohn, M. Lederbauer, S. Mahjoubi, B. Mohr, S.M. Moosavi, A. Naik, A.B. Ozhan, D. Plessers, A. Roy, F. Schöppach, P. Schwaller, C. Terboven, K. Ueltzen, S. Zhu, **J. Janssen**, C. Li, I. Foster, B. Blaiszik. 32 examples of LLM applications in materials science and chemistry: towards automation, assistants, agents, and accelerated scientific discovery. *Machine Learning: Science and Technology*. doi:[10.1088/2632-2153/ae011a](https://doi.org/10.1088/2632-2153/ae011a). Citations: 13
- Code:  [jan-janssen/LangSim](https://github.com/jan-janssen/LangSim)
- J. Janssen**, M.G. Taylor, P. Yang, J. Neugebauer and D. Perez. Executorlib – Up-scaling Python workflows for hierarchical heterogenous high-performance computing. *Journal of Open Source Software*. doi:[10.21105/joss.07782](https://doi.org/10.21105/joss.07782). Citations: 2
- Code:  [pyiron/executorlib](https://github.com/pyiron/executorlib)
- Sk Md A.A. Alvi, **J. Janssen**, D. Khatamsaz, D. Perez, D. Allaire and R. Arróyave. Hierarchical Gaussian process-based Bayesian optimization for materials discovery in high entropy alloy spaces. *Acta Materialia*. doi:[10.1016/j.actamat.2025.120908](https://doi.org/10.1016/j.actamat.2025.120908). Citations: 17
- Code:  [sheikhahnaf/DGP-BO](https://github.com/sheikhahnaf/DGP-BO)

- 2024 **J. Janssen**, E. Makarov, T. Hickel, A.V. Shapeev and J. Neugebauer. Automated optimization and uncertainty quantification of convergence parameters in plane wave density functional theory. *npj Computational Materials*. doi:[10.1038/s41524-024-01388-2](https://doi.org/10.1038/s41524-024-01388-2). Citations: 9
- Code:  [pyiron/pyiron-dft-uncertainty](https://github.com/pyiron/pyiron-dft-uncertainty)
- T.J. Summers, M.G. Taylor, L.J. Augustine, **J. Janssen**, D. Perez, E.R. Batista, and P. Yang. On the Importance of Configuration Search to the Predictivity of Lanthanide Selectivity. *JACS Au*. doi:[10.1021/jacsau.4c00770](https://doi.org/10.1021/jacsau.4c00770). Citations: 14
- S. Menon, Y. Lysogorskiy, A. L. M. Knoll, N. Leimeroth, M. Poul, M. Qamar, **J. Janssen**, M. Mrovec, J. Rohrer, K. Albe, J. Behler, R. Drautz, and J. Neugebauer. From electrons to phase diagrams with classical and machine learning potentials automated workflows for materials science with pyiron. *npj Computational Materials*. doi:[10.1038/s41524-024-01441-0](https://doi.org/10.1038/s41524-024-01441-0). Citations: 22
- Code:  [pyiron/potential_publication](https://github.com/pyiron/potential_publication)
- M. Tynes, M.G. Taylor, **J. Janssen**, D. Burrill, D. Perez, P. Yang, and N. Lubbers. Linear graphlet models for accurate and interpretable cheminformatics. *Digital Discovery*. doi:[10.1039/D4DD00089G](https://doi.org/10.1039/D4DD00089G). Citations: 6
- Code:  [lanl/minervachem](https://github.com/lanl/minervachem)
- 2023 M.G. Taylor, D.J. Burrill, **J. Janssen**, E. Batista, D. Perez, and P. Yang. Architector: high-throughput cross-periodic table 3D complex builder. *Nature Communications*. doi:[10.1038/s41467-023-38169-2](https://doi.org/10.1038/s41467-023-38169-2). Citations: 50
- Code:  [lanl/Architector](https://github.com/lanl/Architector)
- A. Rohskopf, C. Sievers, N. Lubbers, M. A. Cusentino, J. Goff, **J. Janssen**, M. McCarthy, D. Montes de Oca Zapiain, S. Nikolov, K. Sargsyan, E. Sikorski, L. Williams, D. Sema, A. P. Thompson, and M. A. Wood. FitSNAP: Atomistic machine learning in LAMMPS. *Journal of Open Source Software*. doi:[10.21105/joss.05118](https://doi.org/10.21105/joss.05118). Citations: 62
- Code:  [FitSNAP/FitSNAP](https://github.com/FitSNAP/FitSNAP)
- 2021 L.F. Zhu, **J. Janssen**, S. Ishibashi, F. Körmann, B. Grabowski and J. Neugebauer. A fully automated approach to calculate the melting temperature of elemental crystals. *Computational Materials Science*. doi:[10.1016/j.commatsci.2020.110065](https://doi.org/10.1016/j.commatsci.2020.110065). Citations: 49
- Code:  [pyiron/pyiron_meltingpoint](https://github.com/pyiron/pyiron_meltingpoint)
- 2020 T.D. Swinburne, **J. Janssen**, M. Todorova, G. Simpson, P. Plechac, M. Luskin and J. Neugebauer. Anharmonic free energy of lattice vibrations in fcc crystals from a mean field bond. *Physical Review B*. doi:[10.1103/PhysRevB.102.100101](https://doi.org/10.1103/PhysRevB.102.100101). Citations: 9
- Code:  [tomswinburne/BLaSA](https://github.com/tomswinburne/BLaSA)
- 2019 **J. Janssen**, S. Surendralal, Y. Lysogorskiy, M. Todorova, T. Hickel, R. Drautz and J. Neugebauer. pyiron: an integrated development environment for computational materials science. *Computational Materials Science*. doi:[10.1016/j.commatsci.2018.07.043](https://doi.org/10.1016/j.commatsci.2018.07.043). Citations: 176
- Code:  [pyiron/pyiron](https://github.com/pyiron/pyiron)
- Y. Lysogorskiy, T. Hammerschmidt, **J. Janssen**, J. Neugebauer and R. Drautz. Transferability of interatomic potentials for molybdenum and silicon. *Modelling and Simulation in Materials Science and Engineering*. doi:[10.1088/1361-651X/aafd13](https://doi.org/10.1088/1361-651X/aafd13). Citations: 27

- 2016 **J. Janssen**, N. Gunkelmann and H. M. Urbassek. Influence of C concentration on elastic moduli of $\alpha' - Fe_{1-x}C_x$ alloys. *Philosophical Magazine*. doi:[10.1080/14786435.2016.1170224](https://doi.org/10.1080/14786435.2016.1170224). Citations: 13

PUBLICATIONS IN REVIEW

- 2025 Z. Wang, H. Huang, H. Zhao, C. Xu, S. Zhu, **J. Janssen** and V. Viswanathan. DREAMS: Density Functional Theory Based Research Engine for Agentic Materials Simulation. doi:[10.48550/arXiv.2507.14267](https://doi.org/10.48550/arXiv.2507.14267). Citations: 18
- Code:  [BattModels/material_agent](https://github.com/BattModels/material_agent)

OPEN-SOURCE SOFTWARE

- 2015 – present **pyiron** - an integrated development environment for computational materials science
- Role: Lead developer for a team of eight core developers
 - Code:  [pyiron](https://github.com/pyiron)
- 2018 – present **Conda-forge** - community-led software distribution for the conda package manager
- Role: Maintainer for materials science software
 - Contribution: over 1000 packages with a total of over 500 million downloads
 - Code:  [conda-forge/staged-recipes](https://github.com/conda-forge/staged-recipes)
- 2025 **executorlib** - Up-scale python functions for high performance computing (HPC)
- Contribution: Lead the development
 - Code:  [pyiron/executorlib](https://github.com/pyiron/executorlib)
- 2024 **LangSim** - Large language model for atomistic simulation
- Contribution: Lead the team of international researchers
 - Code:  [jan-janssen/LangSim](https://github.com/jan-janssen/LangSim)
- 2022 **Architector** - high-throughput cross-periodic table 3D complex builder
- Contribution: Parallelization of the chemical complex building using mpi4py
 - Code:  [lanl/Architector](https://github.com/lanl/Architector)
- 2021 **FitSNAP** - Software for generating SNAP machine-learning interatomic potentials
- Contribution: Implemented a python library interface for the Exascale computing project
 - Code:  [fitsnap/fitsnap](https://github.com/fitsnap/fitsnap)

WORKSHOPS

- 2025 **Co-organizer** for the workshop on “Machine Learning” as part of the International Max Planck Research School for Sustainable Metallurgy. *ICAMS, Ruhr University*, Bochum, Germany
- 2021 **Co-organizer** for the workshop on “workflows for atomistic simulation”. *Max-Planck-Institut für Eisenforschung*, Düsseldorf, Germany (online)
- Code:  [pyiron-workshop/potentials-workshop-2021](https://github.com/pyiron-workshop/potentials-workshop-2021)
- 2020 **Co-organizer** for the workshop on “software tools from atomistics to phase diagrams”. *Pennsylvania State University*, Pennsylvania, USA (online)
- Code:  [pyiron-workshop/phasediagram-workshop-2020](https://github.com/pyiron-workshop/phasediagram-workshop-2020)

TEACHING

- 2025 **Shared lecture with Prof. G. Sutmann** on “Programming Concepts in Materials Science“ for master students in materials science and simulation. *Ruhr University Bochum*, Bochum, Germany.
- Single lecture** on “From Atoms to Material Properties: An introduction to ab-initio thermodynamics“ as part of the lecture on “Materials Modelling“ for master students in materials science. *Birmingham University*, Birmingham, United Kingdom.
- Single lecture** on “Extension of Density Functional Theory: From Molecules to Solids“ as part of the lecture on “Molecular Photonics and Excited-State Processes (MPESP)“ for master students in chemistry. *Heinrich Heine University*, Düsseldorf, Germany.
- 2024 **Single lecture** on “Density Functional Theory“ as part of the International Max Planck Research School for Sustainable Metallurgy lecture series on Thermodynamics. *Max Planck Institute for Sustainable Materials*, Düsseldorf, Germany
- 2023 **Single lecture** on “Introduction to Materials Informatics“ as part of the Graduate Autumn School of the CRC1349. *Rheinisch-Westfälische Technische Hochschule (RWTH)*, Aachen, Germany

SUPERVISION

- 2025 – present **Prabhath Chilakalapudi** (PhD student): Machine-learned interatomic potentials combining theoretical predictions and experimental references
- Role: Supervisor - acquiring the funding and leading the supervision of the PhD student
- 2024 – present **Haitham Wael Ali Awad Gaafer** (PhD student): Machine-learned interatomic potentials for chemo-mechanics balancing computational efficiency and accuracy
- Role: Supervisor - acquiring the funding and leading the supervision of the PhD student
- Ziqi Wang** (PhD student): Autonomous discovery of novel battery materials with large language model agents and machine-learned interatomic potentials.
- Role: Co-Supervisor - in collaboration with Prof. Viswanathan at University of Michigan
- 2023 – present **Sk Md Ahnaf Akif Alvi** (PhD student): Autonomous in silico processing and materials discovery for additive manufacturing
- Role: Co-Supervisor - in collaboration with Prof. Arroyave at Texas A&M University
- 2023 – 2025 **Ilgar Baghishov** (intern): Multi-fidelity machine-learned interatomic potentials
- Role: Primary Supervisor - acquiring the funding and leading the supervision of the intern
- 2022 – 2023 **Jason Blake Gibson** (intern): Transferability of interatomic machine learning potentials
- Role: Primary Supervisor - acquiring the funding and leading the supervision of the intern
- 2018 **Ankita Biswas** (student assistant): Calculation of vacancy formation energies with pyiron
- Role: Primary Supervisor - for a student research project
- 2017 **Martin Böckmann** (student assistant): Monte Carlo sampling with pyiron
- Role: Primary Supervisor - guiding the student as the first user of the pyiron software

PRESENTATIONS

INVITED TALKS

- 2026 **J. Janssen.** Scalable sustainable materials design with the pyiron workflow framework, *NERSC*, Berkeley, Ca, USA.
- J. Janssen.** Autonomous, Data-Driven Workflows for Materials Acceleration Platforms with pyiron, *DPG Spring Meeting*, Dresden, Germany.
- J. Janssen.** Up-scaling Python functions for HPC with executorlib, *Workflows Community Talks*, Chicago, Il, USA (online).
- 2025 **J. Janssen.** Autonomous Materials Science Research, *European Space Agency AI STAR Symposium*, Darmstadt, Germany.
- J. Janssen.** Can a machine predict novel sustainable materials? Towards using LLMs for Materials Design, *Birmingham University*, Birmingham, United Kingdom.
- J. Janssen.** Autonomous Materials Science Research, *German Research Centre for Artificial Intelligence*, Saarbrücken, Germany.
- J. Janssen.** Up-scale Python Functions for High-Performance Computing with Executorlib, *Department of Energy Python Exchange*, New York, NY, USA (online).
- J. Janssen.** Predict the chemistry of sustainable materials with machine learning, *Seminar in department of Chemistry at the University of Utah*, Salt Lake City, UT, USA.
- J. Janssen.** Accelerating Materials Discovery with Machine Learning, *Nanomaterials: Computation, Theory, Machine Learning and Experiment*, Telluride, CO, USA.
- J. Janssen.** Large Language Model Agents Towards Autonomous Materials Discovery, *Center for Nonlinear Studies (CNLS) Seminar at Los Alamos National Laboratory*, Los Alamos, NM, USA.
- J. Janssen.** Machine Learned Interatomic Potentials for Extreme Environments, *Workshop on Models and Data for Plasma-Material Interactions in Fusion*, Vienna, Austria.
- J. Janssen.** pyiron – Workflows for data-driven Materials Science, *NHR4CES Community Workshop*, Darmstadt, Germany (online).
- J. Janssen.** Predicting Sustainable Materials with Machine Learning, *Machine learning with computational materials science data for modelling nanocrystal catalysts*, Freiburg, Germany.
- 2024 **J. Janssen.** How to use machine learning in the discovery and design of materials for the future?, *Michigan Institute for Computational Discovery & Engineering*, Michigan, MI, USA.
- J. Janssen.** Foundation Models for Materials Discovery, *Summer School at Center for Scientific Foundation Models*, Michigan, MI, USA (online).
- J. Janssen.** pyiron: Simulation Workflows for Data-Driven Materials Design, *Center for Nonlinear Studies (CNLS) Seminar Los Alamos National Laboratory*, Los Alamos, NM, USA.
- J. Janssen.** pyiron: workflows for the development and assessment of interatomic potentials, *Workshop on Machine Learning Force Fields at the institute for mathematical and statistical innovation (IMSI)*, Chicago, Il, USA.

- 2023 **J. Janssen** and J. Neugebauer. Rapid prototyping and up-scaling atomistic workflows with pyiron, *Ab initio Description of Iron and Steel (ADIS): Digitalization and Workflows*, Kreuth, Germany.
- J. Janssen**. Packaging Scientific Software with Conda-Forge, *Workshop at the Institute for Pure and Applied Math (IPAM)*, Los Angeles, CA, USA.
- J. Janssen** and D. Perez. Up-scaling atomistic simulation workflows with pyiron, *Workshop at the Institute for Pure and Applied Math (IPAM)*, Los Angeles, CA, USA.
- J. Janssen**, M. G. Taylor, P. Yang and D. Perez. Tutorial on high-throughput screening for chemical science with pyiron, *2nd International Workshop on Theory Frontiers in Actinide Science : Chemistry & Materials*, Santa Fe, NM, USA.
- 2022 **J. Janssen**, T. Hickel and J. Neugebauer. Automated atomistic calculation of thermodynamic and thermophysical data, *DPG Fall Meeting*, Regensburg, Germany.
- J. Janssen**. pyiron - an integrated development environment (IDE) for scientific workflows at scale, *Department of Energy Python Exchange*, New York, NY, USA (online).
- J. Janssen**. Up-scaling simulation protocols with pyiron, *Platform Material Digital Workflow Meeting 2022*, Berlin, Germany (online).
- J. Janssen**. pyiron - an integrated development environment (IDE) for materials science, *Special Interest Group Data Infrastructure (SIGDIUS) Seminar*, Stuttgart, Germany (online).
- 2021 **J. Janssen** and J. Neugebauer. pyiron - an integrated development environment for materials science, *CECAM Workshop: Simulation Workflows in Materials Modelling (SWiMM)*, Lausanne, Switzerland (online).
- 2020 **J. Janssen**, T. Hickel and J. Neugebauer. Uncertainty quantification for *ab initio* thermodynamics, *Group Seminar Professor R. Drautz*, Bochum, Germany (online).
- J. Janssen**, T. Hickel and J. Neugebauer. Automated *ab initio* determination of materials properties at finite temperatures with pyiron, *NIST Workshop: Atomistic simulations for industrial needs*, Gaithersburg, MD, USA (online).
- 2019 **J. Janssen**, T. Hickel and J. Neugebauer. Automated *ab initio* determination of materials properties at finite temperatures with pyiron, *Center for Nonlinear Studies (CNLS) Seminar at Los Alamos National Laboratory*, Los Alamos, NM, USA.
- J. Janssen**, T. Hickel and J. Neugebauer. pyiron - an integrated development environment for computational materials science, *Group Seminar Professor G. Kresse*, Vienna, Austria.

TALKS AT INTERNATIONAL CONFERENCES

- 2026 T. Hickel, **J. Janssen** and J. Neugebauer. Data-Driven and LLM-Based Design of Hydrogen Solubilities in Metallic Alloys, *TMS Spring Meeting*, San Diego, Ca, USA.
- J. Janssen** and J. Neugebauer. Large Language Model Agents for Atomistic Simulation Workflows, *TMS Spring Meeting*, San Diego, Ca, USA.
- J. Janssen**, P. Chilakalapudi, H. Gaafer and J. Neugebauer. Uncertainty Propagation for Ab Initio Thermodynamic Phase Diagrams, *TMS Spring Meeting*, San Diego, Ca, USA.

H. L. Mai, **J. Janssen**, T. Hickel and J. Neugebauer. Accelerating grain boundary segregation studies in ferritic steels with machine learned interatomic potentials, *DPG Spring Meeting*, Dresden, Germany.

J. Janssen. Leveraging AI in Simulation Workflows for Sustainable Materials Design with pyiron, *deRSE Conference*, Stuttgart, Germany.

2025 **J. Janssen**. Up-scaling Python functions for HPC with executorlib, *Supercomputing Conference*, St. Louis, MO, USA.

J. Janssen and J. Neugebauer. Automated Workflows for Fitting Machine Learned Interatomic Potentials on the Exascale with pyiron, *Psi-k conference*, Lausanne, Switzerland.

J. Janssen and J. Neugebauer. Large Language Model Agents for Atomistic Simulation Workflows, *MSE Research Data Forum*, Siegburg, Germany.

J. Janssen and J. Neugebauer. Inverse Materials Design with Large Language Models, *DPG Spring Meeting*, Regensburg, Germany.

2024 **J. Janssen**, J. Neugebauer and D. Perez. Potential in a day - up-scale the fitting of machine learning potentials for the Exascale, *IPAM Workshop*, Los Angeles, CA, USA.

J. Janssen and J. Neugebauer. pyiron - Simulation Workflows for the Design of Sustainable Materials, *MRS Fall Meeting*, Boston, MA, USA.

M. Stricker, L. Banko, N. Sarazin, **J. Janssen**, N. Siemer, J. Neugebauer and A. Ludwig. Computationally Accelerated Experimental Materials Characterization Using pyiron, *Multiscale Materials Modelling (MMM) conference*, Prague, CZ.

J. Janssen and J. Neugebauer. Automated Workflows to Construct and Validate Machine-Learned Interatomic Potentials, *Multiscale Materials Modelling (MMM) conference*, Prague, CZ.

I. Baghishov, **J. Janssen**, G. Henkelman and D. Perez. Investigating the uncertainty in multi-fidelity machine learning interatomic potentials, *TMS Spring Meeting*, Orlando, FL, USA.

J. Janssen and D. Perez. Transferable Machine Learning Potentials for Extreme Environments, *TMS Spring Meeting*, Orlando, FL, USA.

2023 **J. Janssen** and D. Perez. Parameter studies for interatomic potentials using LAMMPS and pyiron, *LAMMPS Virtual Workshop and Symposium 2023*, Philadelphia, PA, USA (online).

J. Janssen and D. Perez. Enabling long timescale molecular dynamics simulation with ab initio precision, *MRS Spring Meeting*, San Francisco, CA, USA.

J. Janssen and D. Perez. pyiron an integrated development environment for the development and assessment of interatomic models, *11th Annual Mach Conference*, Baltimore, MD, USA.

J. Janssen and D. Perez. Enabling long timescale molecular dynamics simulation with ab initio precision, *TMS Spring Meeting*, San Diego, CA, USA.

2022 **J. Janssen**, T. Hickel and J. Neugebauer. Predicting melting temperatures from bulk properties with pyiron, *Multiscale Materials Modelling (MMM) conference*, Baltimore, MD, USA.

J. Janssen, M. G. Taylor, P. Yang and D. Perez. Screening of ligand-metal-complexes for separation science with pyiron, *ACS Fall Meeting*, Chicago, IL, USA.

J. Janssen. pyiron – an integrated development environment (IDE) for scientific workflows, *Scientific Computing with Python (SciPy)*, Austin, TX, USA.

J. Janssen. Predicting melting temperatures from bulk properties with pyiron, *Artificial Intelligence for Materials Science (AIMS)*, Gaithersburg, MD, USA (online).

2019 **J. Janssen**, T. Hickel and J. Neugebauer. Automated uncertainty quantification for *ab initio* thermodynamics, *MRS Fall Meeting*, Boston, MA, USA.

J. Janssen, T. Hickel and J. Neugebauer. Automated sensitivity analysis for high-throughput *ab initio* calculations, *IPAM Workshop*, Los Angeles, CA, USA.

J. Janssen, T. Hickel and J. Neugebauer. Automated error analysis and control for *ab initio* calculations, *DPG Spring Meeting*, Regensburg, Germany.

J. Janssen, T. Hickel and J. Neugebauer. Automated sensitivity analysis for high-throughput *ab initio* calculations, *TMS Spring Meeting*, San Antonio, TX, USA.

2018 **J. Janssen**, T. Hickel and J. Neugebauer. Generation of *ab initio* datasets with predefined precision using uncertainty quantification, *DPG Spring Meeting*, Berlin, Germany.

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